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# Joint angle measurement: a comparative study of the reliability of goniometry and wire tracing for the hand

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**Objectives:** To compare the inter- and intra-rater reliability of goniometry and wire tracing in the assessment of finger joint angles: metacarpo-phalangeal (MCPJ), proximal (PIPJ) and distal interphalangeal joints (DIPJ).
**Design:** Twenty occupational therapists and 20 physiotherapists with a range of clinical experience were recruited from nine different centres. Using a masked goniometer and wire tracing they carried out repeated assessments of the MCPJ, PIPJ and DIPJ of a normal subject fixed in two different positions.
**Results:** The two assessment methods did not produce comparable angle measurements. Goniometry showed greater inter- and intra-rater reliability than wire tracing. Regardless of the assessment tool, the repeatability coefficient indicated that DIPJ measurement was less reliable than the other joints. Clinical and specialist experience did not affect reliability.
**Conclusion:** Although both goniometry and wire tracing show limitations as reliable assessment tools, it is recommended that where possible goniometry should be used.

## Introduction

The range of motion (ROM) at a joint is regarded by many clinicians as a measurable and definable entity and is therefore frequently used as an assessment tool. Joint angle measures are also used as an index for change and as an outcome measure to evaluate the effects of medical, surgical and other therapeutic interventions.

Traditional methods of measurement of hand ROM include visual estimation, goniometry, composite finger flexion to distal palmar crease, torque range of motion and wiregrams (wire tracing).[1–3] Wire tracing refers to the process whereby a solder wire is laid along the dorsal aspect of the hand and bent to conform to the angle of each joint. The wire is then lifted off the finger and laid on paper, where its shape is traced with a pencil. In 1986 the British Orthopaedic Association, in conjunction with the British Society for Surgery of the Hand, recommended a hand assessment chart that included the use of both goniometry and wire tracing for the measurement of joint motion angles.[4]

Any measurements that are used as a basis for treatment planning or for research purposes need to be reliable. Reliability of joint measurement refers to the amount of agreement between successive measures of the same joint, the greater

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the agreement the more reliable the measure.[5] Intra-rater reliability examines the ability of a single tester to obtain consistent results on successive measures; inter-rater reliability refers to the amount of agreement between measurements for the same joint by different testers.

A number of studies have examined the reliability of goniometry using specific instrumentation, for specific joints and under specific conditions.[6–15] Some studies have compared different types of goniometer,[16–21] while others have compared goniometry with other instrumentation.[22–25] Comparisons of goniometry with visual estimation have been carried out by a number of researchers.[26–29] Extensive general reviews of goniometry have been written,[30–34] and issues of reliability have also been examined.[35–37]

No published studies to date have compared goniometry with wire tracing for assessment of finger joint angles. The general consensus seems to be that Salter[34] was correct in her assertion in 1955 that goniometers are very accurate and reliable, and that inaccurate measurements are the result of faulty application. Despite the potential for error, many authors have found goniometry to have high intra-rater reliability: Low[26] found the mean intra-rater error of five testers measuring elbow flexion was less than 5 degrees over 10 measurements, and Hellebrandt *et al.*[37] found that a skilled observer varied 7 degrees or less 95% of the time. Inter-rater reliability has generally been found to be lower. Previous work examining goniometry for finger joint angles carried out by Hamilton and Lachenbruch[17] used seven skilled testers to assess the reliability of three different goniometers measuring metacarpo-phalangeal joint (MCPJ), proximal interphalangeal joint (PIPJ), and distal interphalangeal (DIPJ) joints of the hand, lateral to the joint or dorsal to the joint. This study showed small variance for inter-rater reliability but no significant difference in results among the three different types of goniometer.

A thorough literature search did not reveal any published studies about the reliability of wire tracing as a method of assessment of ROM.

In view of the lack of data on the reliability of joint measures of the hand, the aims of this study were:

- to determine the intra-rater and inter-rater reliability for goniometry and wire tracing measurement of the MCPJ, PIPJ, and DIPJ of the hand; and
- to compare the reliability of these two different methods.

## Method

### Subject

A healthy subject with no known existing pathology had the middle finger of each hand fixed in two different positions using a removable splint made of thermoplastic material (Rolyan Ezeform, Smith & Nephew, Hull, UK). These two positions reproduced common clinical limitations of movement: position 1 with all joints in some flexion, position 2 with some limitation of full extension at the MCPJ and PIPJ but full hyperextension of the DIPJ (Figure 1).

Forty volunteers were recruited from nine different sites, four of which specialized in hand therapy. Volunteers were qualified physiotherapists (20) and occupational therapists (20) with clinical experience ranging from < 1 to 16+ years, 25 (62.5%) of them having at least one year's specialist experience in hand therapy. All therapists completed a short questionnaire indicating the method(s) they routinely used to assess ROM of the hand (methods of measurement questionnaire).

### Instrumentation

The goniometer chosen for the study was the Rolyan Flexion/hyper-extension Finger Goniometer[38] (Smith & Nephew, Hull, UK). This has one short arm with a protractor head and one long arm to facilitate placement over the small joints of the hand, it permits measurement from 30 degrees of hyper-extension to 120 degrees of flexion. The goniometer scale measures in 1 degree increments, this was covered with contact paper so the testers could not read the scale but the recorder could read them from the reverse side. As previous researchers[17] found no significant variance in results among the three different types of goniometer the same goniometer was used throughout the study. The wire used was 16 SWG gauge tin/lead solder wire, 25 cm long,

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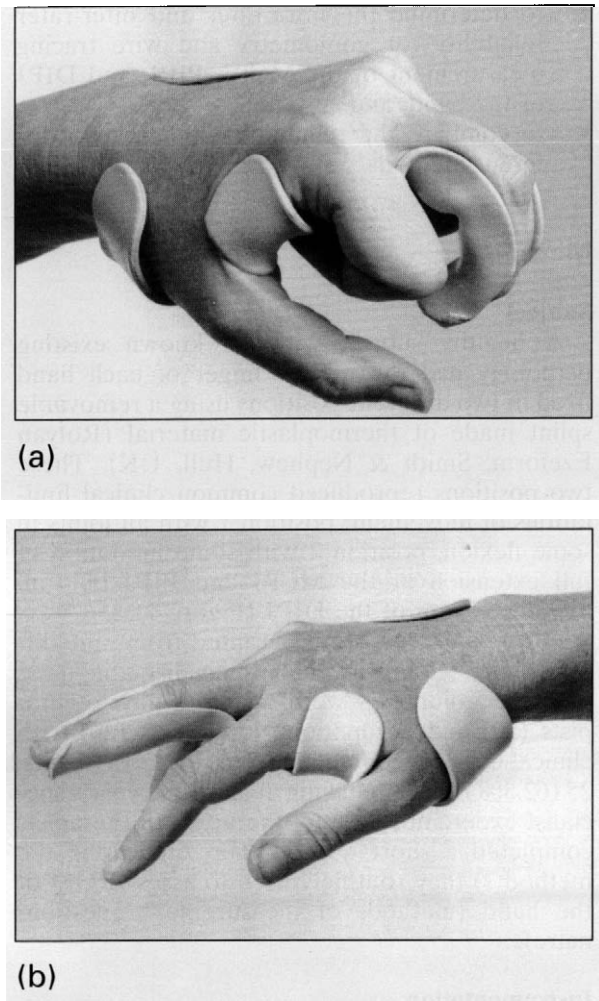



**Figure 1** (a) Subject's hand splinted in position 1. (b) Subject's hand splinted in position 2.

inserted into 2 mm silicone rubber tubing as recommended in the literature.[1,2] Six pieces of wire were supplied to enable the tester to use a fresh piece of straight wire for each tracing.

## Procedure

Therapists agreeing to participate in the study received a protocol for wire tracing and goniometry prior to the day of testing. This allowed those unfamiliar with either measurement approach to familiarize themselves with the methods of measurement, the lack of a measurement protocol available prior to testing perhaps having attributed to poor results in some previous studies.[3,26] Both measures were taken on the dorsal aspect of the joint. Dorsal measurement with the goniometer was chosen as this is one of the recommended methods of measurement,[3] has been demonstrated to be as reliable as lateral placement,[17] and provided data comparable with wire tracing. The wrist and forearm were held in a standard manner for each test.

After completing the methods of measurement questionnaire, each therapist measured the MCPJ, PIPJ and DIPJ of position 1, aligning the arms of the goniometer to their satisfaction. When they were satisfied with the placement, the recorder read the value. The therapist then performed a wire tracing of the same position, transcribing the wire tracing to paper as described in the protocol. The therapist was asked to identify each joint by a single line bisecting the tracing. This process was repeated for position 2. The whole procedure was carried out three times. Thus a total of six measures of six joints was taken by each therapist. At no point during the measurement phase were therapists permitted to view their results. A short time interval elapsed between each joint measurement method repetition, while the alternate method and other joints were measured. Previous research indicated[36] that intra-rater reliability should be tested with short time intervals only as this provides a more accurate evaluation, reducing uncontrolled variables.

The same recorder was used at each occasion for recording all goniometric measurements. Her accuracy was tested for reading the goniometer and was found to give a maximum error of one degree. A different recorder transcribed all the wire tracing angles using a protractor, adhering to a strict protocol. This ensured there was no bias and allowed standardized measurement.

## Analysis of data

The recorded measurements and observer characteristics were databased in SPSS, and later transferred to SAS for statistical analysis using the SAS/STAT procedures VARCOMP and GLM. The measurements were initially summarized using descriptive statistics, with particular focus being placed on the ranges and variances. Subsequently the variance for each measurement

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was separated into components due to between-observer and within-observer variation. Repeatability coefficients for the same and different observers were calculated from these. A repeatability coefficient for the same observer estimates the 95% limit on the difference between two measurements made by the same therapists. We can expect any pair of measurements to be within this distance of each other, 95% of the time. Similarly, a repeatability coefficient for different observers estimates the 95% limit on the difference between two measurements made by different therapists.

Differences in means with respect to occupation, years of clinical and specialist experience, routine use and experience of measurement tools were assessed using a repeated measures analysis of variance. Differences in within-observer variances with respect to these subgroups were assessed using an *F*-test for the ratio of two variances.

## Results

The results of the methods of measurement questionnaire completed by the therapists are displayed in Table 1. Measurement of joint angles was most commonly performed in clinical practice using goniometry. Although the use of wire tracing seems much lower it must be noted that half the volunteers did not commonly assess hands. As wire tracing is of value only in the measurement of small joints, its use is probably limited to clinicians treating hands. Although all participants had received training in the principal of goniometry, familiarity with the use of wire tracing was limited to 22 (55%) participants. Those who were familiar with the use of wire tracing had acquired this skill in a number of different ways including training schools, in-service education postgraduate courses and from other colleagues. One participant was self-taught.

**Table 1** Routine method of hand joint assessment

| Method of assessment | Responses |
|---|---|
| Goniometry alone | 20 (50%) |
| Visual estimation alone | 8 (20%) |
| Wire tracing alone | 3 (7.5%) |
| Wire tracing + visual estimation | 2 (5%) |
| Wire tracing + goniometry | 1 (2.5%) |
| Goniometry + visual estimation | 5 (12.5%) |
| Wire tracing + goniometry + visual estimation | 1 (2.5%) |

Table 2 shows the means of the three measures of the same joint with the standard deviations and ranges. In a study such as this, where there is no gold standard for reference, the variances, standard deviations and ranges of the results are of greater importance than the means. As can be seen, goniometry consistently produced smaller ranges and lower standard deviations than wire tracing. Interestingly, for both measures there was a tendency for ranges and standard deviations to increase with the more distal joints. This

**Table 2** Descriptive statistics of goniometry and wire tracing for positions 1 and 2

| Joint | Method | Mean | SD | Minimum | Maximum | Range |
|---|---|---|---|---|---|---|
| *Position 1* | | | | | | |
| MCP | Goniometry | 44.46° | 1.09 | **41°** | 47° | 6° |
| | Wire tracing | 41.58° | 2.58 | **36°** | 47° | 11° |
| PIP | Goniometry | 74.93° | 2.16 | **70°** | 79° | 9° |
| | Wire tracing | 70.40° | 2.98 | **61°** | 76° | 15° |
| DIP | Goniometry | 47.46° | 2.94 | **42°** | 55° | 13° |
| | Wire tracing | 42.13° | 3.14 | **37°** | 49° | 12° |
| *Position 2* | | | | | | |
| MCP | Goniometry | 7.76° | 1.68 | **4°** | 13° | 9° |
| | Wire tracing | 4.38° | 2.84 | **–2°** | 9° | 11° |
| PIP | Goniometry | 14.67° | 1.70 | **10°** | 17° | 7° |
| | Wire tracing | 10.88° | 2.42 | **6°** | 17° | 11° |
| DIP | Goniometry | –1.61° | 3.04 | **–9°** | 6° | 15° |
| | Wire tracing | –3.10° | 3.55 | –10° | 5° | 15° |

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would suggest that goniometry is generally more reliable than wire tracing, but that the distal joints with shorter levers create more difficulty in measurement using either method. In the clinical situation, therefore, any alteration found in the range of the more distal joints should be interpreted with caution.

In Table 3 the variance of the 120 observations for a given response was divided into the components of variance attributable to intra- and inter-rater variation. The repeatability coefficient (RC) is the 95% limit on the difference between two repeated measurements on the same subject, i.e. 95% of the time, any two measures will be within this range of each other. Thus the lower the RC, the greater the reliability. As can be seen, the RCs are considerably lower for goniometry than wire tracing. The higher RCs for measurement of the more distal joints when both wire tracing and goniometry were used indicates the greater reliability of measurement of the MCP joint.

No significant differences in means were found with respect to occupation, years of clinical and specialist experience, routine use or experience of measurement tools. For certain hand positions and joints, differences in within-observer variances were found to be due to some of these factors, although there was no single subgroup which showed greater reliability (i.e. a smaller variance) across the range of joints and positions. The reliability of therapists who were familiar with either measurement tool was no greater than those with no such familiarity.

## Discussion

These results support the finding from previous studies that when assessing ROM, inter-rater reliability is considerably poorer than intra-rater reliability, irrespective of the measurement tool.[6–15,26–29]

In this study, goniometry is shown to have consistently better reliability than wire tracing with repeatability coefficients in some cases half those of wire tracing. The goniometry results for intra-rater reliability compare favourably with the findings of Low,[26] but show greater variability than Hamilton and Lachenbruch[17] which may be explained by a number of factors. In the latter study only seven skilled observers from the same place of work were used, in this study 40 observers were used from nine sites with varying experience in joint measurement.

There could be a number of explanations for the greater variability of wire tracing. As nearly half of the participants were not familiar with wire tracing, and all were familiar with goniometry, this might have been expected to influence the results; however this did not prove to be the case. There would seem to be a greater potential for error inherent in the use of the wire, for example as a result of distorting the wire in the

**Table 3** Variance components and repeatability coefficients

| | | Position 1 | | | Position 2 | | |
|---|---|---|---|---|---|---|---|
| Method | Measure | MCP | PIP | DIP | MCP | PIP | DIP |
| Goniometer | $\sigma^2$ between observers | 0.59 | 3.78 | 6.93 | 2.05 | 2.10 | 7.73 |
| | $\sigma^2$ within observers | **1.80** | **2.7** | 5.07 | 2.31 | 2.33 | 4.61 |
| | *Repeatability coefficient:* | | | | | | |
| | Same observer | 3.8° | 4.7° | 6.4° | 4.3° | 4.3° | 6.1° |
| | Different observers | **4.4°** | **7.2°** | 9.8° | 5.9° | 6.0° | 9.9° |
| Wire tracing | $\sigma^2$ between observers | **3.31** | 3.95 | 5.88 | 5.37 | 3.11 | 7.99 |
| | $\sigma^2$ within observers | **9.97** | 14.86 | 11.95 | 8.08 | 8.23 | 13.80 |
| | *Repeatability coefficient:* | | | | | | |
| | Same observer | 8.9° | 10.9° | 9.8° | 8.0° | 8.1° | 10.5° |
| | Different observers | 10.3° | 12.3° | 11.9° | 10.4° | 9.5° | 13.2° |

The repeatability coefficient is the 95% limit on the difference between two measurements.

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removal from the subject's hand and/or in tracing the outline on to the paper, in the orientation of the wire on the paper, and in the correct identification of the joints on the traced line. Additional sources of error could include the recorder's interpretation of the tracing and lack of consistency in the therapist's technique. As only one type of wire was used, a further study might be of value to assess the reliability of different gauges and types of wire.

The potential for error observed in the use of goniometry include: variation in the amount of pressure placed on the arms of the goniometer by therapists; interference with the accurate alignment of the arms of the goniometer by the therapists' fingers; error in the identification of the anatomical landmarks, such as the DIPJ line; and lack of consistency in therapist technique.

As the purpose of this study was to measure the reliability of therapists using specific measurement tools, the subject's hands were normal, with no oedema or wounds to complicate the procedure. In a clinical setting these might alter the ability to use different methods of joint angle measurement and the reliability of measures taken.

An additional important finding not previously demonstrated was the considerable variation in the reliability of measurement of different joints of the finger. The measurement of the MCPJ consistently showed greater repeatability than the more distal joints, with the DIPJ producing greatest inconsistencies. This can be explained in part by the much shorter lever arms to align the wire or goniometer, less definable joint line (particularly in hyperextension) and lack of a distal bony point or landmark for alignment. This is of importance when assessing reported outcomes using joint angle measurements following medical or surgical interventions, for example the results of surgical repair of the long flexor tendons serving the distal inter-phalangeal joints.

This study highlights potential problems with the use of joint angle measures to evaluate response to surgical intervention or treatment, for example the rating of ROM outcome following repair of flexor tendons. Such results are usually calculated as the alteration in joint angles achieved, for example the calculation of the total combined ROM of DIPJ and PIPJ as a percentage of the anticipated normal ROM.[38] The results of this study would indicate that using comparative measurements obtained from wire tracing might be a less reliable outcome measure.

## Conclusion

Joint angle measures are widely used to provide baseline information for comparison with subsequent measures to indicate change and therefore outcome. Although wire tracing can undoubtedly indicate trends in change, in view of the wide variation both within-observer and between-observer, this would seem to be its principal value. Goniometry provided greater reliability in both within- and between-observer testing indicating that, where possible, this should be the measurement tool of choice.

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